

SOC Internship - Day 6

1. What is SIEM?

SIEM stands for **Security Information and Event Management**.

It is a cybersecurity system that helps organizations **collect, analyze, and monitor security data** from different sources in real time.

2.Key Functions of SIEM

Log Collection

- Gathers logs from multiple sources (Linux, Windows, network devices)

Event Correlation

- Connects different events to detect attacks
(*Example: multiple failed logins → possible brute force attack*)

Real-time Monitoring

- Continuously watches system activity

Alerting

- Sends alerts when suspicious behavior is detected

Reporting & Compliance

- Generates reports for audits and security standards

3.How SIEM Works

1. Log Collection

SIEM gathers logs from:

- Servers
- Firewalls
- Routers
- Antivirus software
- Applications
- Operating systems

2. Log Normalization

Different devices create logs in different formats.

SIEM converts all logs into a standard format for easier analysis.

Example:

- Windows log format
- Linux syslog format
- Firewall log format

All become unified inside the SIEM

3. Event Correlation

SIEM compares and connects multiple events to identify suspicious activity.

Example:

If SIEM detects:

- Many failed login attempts
- Same IP address
- Short time period

It identifies a possible:

- **Brute Force Attack**

4. Threat Detection

SIEM uses:

- Rules
- Signatures
- Behavioral analysis

to detect threats such as:

- Malware
- Phishing
- Unauthorized access
- DDoS attacks

5. Alert Generation

When suspicious behavior is detected, SIEM generates alerts.

Severity: High

Attack: Brute Force

IP: 192.168.1.10

6. Investigation

SOC analysts investigate:

- Logs
- Source IPs
- User activity
- System behavior

to confirm whether it is a real attack.

7. Incident Response

After confirmation, security teams take action:

- Block IP addresses
- Disable accounts
- Isolate infected systems
- Remove malware

SIEM Workflow

Devices & Applications

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Log Collection

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Normalization

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Event Correlation

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Threat Detection

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Alert Creation

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SOC Investigation

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Incident Response

Real-World Example of SIEM

Scenario: Brute Force Attack Detection

A company's server is being targeted by an attacker trying to guess a user's password.

The attacker attempts many logins from the same IP address.

Apr 10 10:17:45 Failed password for admin from 192.168.1.50

Apr 10 10:17:50 Failed password for admin from 192.168.1.50

Apr 10 10:17:55 Failed password for admin from 192.168.1.50

Apr 10 10:18:01 Failed password for admin from 192.168.1.50

Apr 10 10:18:05 Failed password for admin from 192.168.1.50

◆ What SIEM Does

1. Collects Logs

The SIEM receives these logs from the Linux server.

2. Correlates Events

SIEM notices:

- Multiple failed logins
- Same username
- Same IP address
- Very short time interval

3. Detects Threat

Alert: Possible Brute Force Attack

Severity: High

Target User: admin

Source IP: 192.168.1.50

5. SOC Analyst Investigates

The analyst checks:

- Whether the IP is trusted
- Login history
- Geolocation
- Previous alerts

6. Response Actions

The security team may:

- Block the IP address
- Lock the account temporarily
- Enable MFA → **Multi-Factor Authentication**
- Investigate further activity

Popular SIEM Tools

Splunk

ELK Stack (Elasticsearch, Logstash, Kibana)

IBM Qradar

Class Activity:

Draw a SIEM workflow diagram showing log collection, analysis, and alerting

